Adaptive Intent-Preference Arbitration for Conversational Recommendation

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***Abstract*—A conversational recommender hears two voices at once: what a user has liked over many sessions, and what they are asking for in the current conversation. Most systems blend the two with a fixed rule and hope for the best. That works when the signals agree and fails quietly when they do not, for example when a habitual comedy viewer asks for a war film for the weekend. We propose *Adaptive Intent–Preference Arbitration* (AIPA), a model that first classifies the relationship between the long-term preference (LTP) and the short-term intent (STI) into one of five categories (complement, consistent, conflict, override, uncertain), then chooses an action (fuse, prioritise LTP, prioritise STI, or ask a clarifying question), and finally re-ranks candidates with learned, action-conditioned fusion weights. Two further components track whether an intent shift repeats across sessions and should be absorbed into the long-term profile, and probe the trained model with counterfactual ablations of each signal to report which one actually drove a recommendation. We build the whole pipeline on ReDial, an English human–human movie recommendation corpus, deriving cross-session profiles from recurring seeker identities and using MovieLens genres as item metadata. Because ReDial has no relationship annotations, labels come from a transparent weak-rule scheme and from controlled synthetic injection of all five relationship classes, and every result is reported separately for the two sources. Training emphasises disagreement turns through conflict-weighted losses, relationship label smoothing and confidence-filtered self-training, with every choice made on the validation split. On the full corpus with 5 seeds and a pre-trained sentence encoder, AIPA is far better than profile-only, GRU and SASRec baselines, but it stays about one point of Hit@10 behind an intent-only model and a KBRD-style conversational baseline, on all turns and on the conflict subsets alike, and the ablations are statistically flat. The parts of the model that are meant to make its decisions legible do work: the relationship head reaches macro-F1 0.63 against the weak-rule reference, clarification precision is 0.99 and wrong overrides are about 1%. We report these outcomes as they are, discuss why the central hypothesis is not supported on this corpus, and list the concrete steps needed for a conclusive test. All numbers, tables and figures in this paper are generated by the released code.**

***Index Terms*—Conversational recommender systems, user intent, long-term preference, preference conflict, clarification, counterfactual analysis, reproducibility.**

I. Introduction

RECOMMENDER systems that talk to their users have an advantage that click-based systems lack: the user can simply say what they want right now. The flip side is that this fresh signal has to be reconciled with everything the system already knows about the person. Someone who has spent years watching dramas may ask, on a Friday night, for "something dumb and funny". A system that follows its profile ignores the request; a system that follows the request may throw away a decade of evidence for a single evening. Neither extreme is right, and the right answer depends on the situation.

The bulk of work on conversational recommendation (CRS) has focused on how to elicit preferences within a dialogue [1]–[4] or how to ground dialogue in knowledge about items [5], [6]. How to reconcile what is said now with what was learned before has received much less direct attention. When both signals are used, they are usually combined with a fixed rule or a learned gate that is never asked to explain itself, and evaluation almost never singles out the turns where the two disagree. Yet those are exactly the turns where the design choice matters.

This paper treats the reconciliation step as a first-class problem. We call it *arbitration*, and we make it explicit in three ways. First, the model predicts what kind of relationship holds between the long-term preference (LTP) and the short-term intent (STI): they may reinforce each other, one may add a new dimension the other lacks, they may contradict each other, the user may explicitly set the profile aside, or there may simply be too little evidence to tell. Second, this relationship drives a discrete action (fuse the signals, favour one of them, or ask a question) and the action in turn sets the fusion weights. Third, the model is made to account for its recommendations: after training we knock out one signal at a time and measure how much the ranking moves, giving a per-turn label of which signal drove the recommendation. We stress that this is a diagnostic of the fitted model, not a causal claim about users.

Our second aim is a clean and fully reproducible experimental base. We deliberately use ReDial [7], an English corpus of human–human movie dialogues. It has no cross-session profiles by design, but crowd workers recur across many dialogues, and we use that recurrence to construct chronological seeker histories under strict no-look-ahead rules. ReDial also has no labels for our relationship taxonomy; rather than pretend otherwise, we derive weak-rule labels from the dialogue text and additionally inject controlled synthetic conflicts, and we never mix the two in a table. Every claim about a metric is accompanied by bootstrap intervals and paired tests with multiple-comparison correction.

Code, configuration, generated figures and the executed notebook are available at https://github.com/dhayarajas/AIPA-CRS.

The results of the run reported here are mixed, and we say so. AIPA improves on profile-only and sequential baselines (GRU, SASRec) by a clear margin, matches a KBRD-style conversational baseline, its arbitration behaviour is sensible, and on controlled conflicts it follows the stated intent. It does not, however, outperform an intent-only recommender, which is in fact slightly and significantly better on the natural test set, and none of the ablations reach significance. We see the contribution of this paper as the framing, the evaluation protocol, and an honest baseline measurement that later work can build on; the discussion in Section VIII lists what a decisive test would need.

II. Related Work

A. Conversational recommendation

Early CRS work framed the dialogue as a preference elicitation game in which the system asks about attributes and the user answers [1], [2]. Estimation–Action–Reflection [3] formalised the choice between asking and recommending as a policy problem. The "system ask, user respond" paradigm of Zhang *et al.* [4] likewise treats clarifying questions as an action to be planned. Our clarification action sits in this tradition but is triggered by the relationship between two evidence sources rather than by attribute uncertainty alone.

Open-ended, natural-language CRS became tractable with ReDial [7], followed by knowledge-grounded models such as KBRD [5] and KGSF [6], and by further corpora including INSPIRED [8] and the Chinese TG-ReDial [9]. TG-ReDial is the only one of these that ships per-user histories, but its dialogues are in Chinese; we chose ReDial so that data, labels, case studies and error analysis could all be read in English, and reconstructed histories from recurring seekers instead. Surveys by Jannach *et al.* [10] and Gao *et al.* [11] give a broader map of the field; both note that reconciling the conversation with the historical profile remains open.

B. Long- and short-term signals

Session-based and sequence-aware recommenders [12]–[14] learn to weigh recent behaviour against older behaviour implicitly, through the recurrence or the attention pattern. Our sequential baseline follows this recipe. The difference in AIPA is that the weighing is exposed: the relationship class and the action are discrete, supervised (weakly) and inspectable, and the fusion weights are a function of them.

C. Counterfactual reasoning in recommendation

We borrow the vocabulary of interventions [15] for the driver diagnostic, but the object we intervene on is the trained model, not the user. Zeroing the LTP encoding and re-scoring tells us how much the model's output depends on that pathway, which is a form of ablation-based attribution. We do not claim identification of any causal effect on user behaviour, and we avoid the word "effect" when describing these numbers.

TABLE I
NOTATION.

Symbol	Meaning
$\mathcal{I}$, $\mathcal{G}$	item catalogue; genre set ($\lvert\mathcal{G}\rvert = 19$)
$G(i) \subseteq \mathcal{G}$	MovieLens genres of item i (may be empty)
u, c, t	seeker, dialogue (conversation id), turn index
$\mathcal{H}_u(c)$	dialogues of u strictly before c, Eq. (1)
$\mathbf{h} = (h_1, \ldots, h_L)$	liked/seen items from $\mathcal{H}_u(c)$, $L \le L_{\max}$
$\mathcal{P}$	profile sentences of u from $\mathcal{H}_u(c)$
R_t, ℓ_t	recent seeker text and last seeker utterance before t
$\mathcal{M}_{<t}$, $\mathcal{M}^{+}_{<t}$	movies mentioned / liked by the seeker before t
$\boldsymbol{\pi}_{\mathrm{L}}, \boldsymbol{\pi}_{\mathrm{S}} \in \bar{\Delta}^{\lvert\mathcal{G}\rvert}$	LTP and STI genre vectors, Eqs. (3), (4)
$\mathbf{f} \in \mathbb{R}^8$	lexical and length flags, Eq. (5)
$\phi(\cdot) \in \mathbb{R}^{d_t}$	text encoder, Eq. (6)
$\mathbf{e}_i$, b_i	item vector and item bias, Eq. (7)
$\mathbf{h}_{\mathrm{L}}, \mathbf{h}_{\mathrm{S}} \in \mathbb{R}^d$	LTP and STI encodings
$\mathbf{s}_{\mathrm{L}}, \mathbf{s}_{\mathrm{S}}, \mathbf{s} \in \mathbb{R}^{\lvert\mathcal{I}\rvert}$	component and fused score vectors
$\mathcal{R}$, $\mathcal{A}$	relationship classes, actions ($\lvert\mathcal{R}\rvert = 5$, $\lvert\mathcal{A}\rvert = 4$)
$\mathbf{p} \in \Delta^5$, $\mathbf{q} \in \Delta^4$	relationship posterior, action distribution
$\mathbf{W}_A \in \mathbb{R}^{4\times 2}$	action-to-weight table (learned), Eq. (17)
$\mathbf{c} \in \mathbb{R}^5$	counterfactual features, Eq. (23)
$\gamma, \theta, \tau, \rho, k, \eta$	decay, rule threshold, driver threshold, dominance ratio, persistence count, persistence gain

III. Problem Formulation

This section fixes notation and states the prediction problem exactly as the released code constructs it. Table I collects the symbols. Vectors are bold lower case ($\mathbf{h}$), matrices bold upper case ($\mathbf{W}$), sets calligraphic ($\mathcal{I}$), and $\mathbb{1}[\cdot]$ is the indicator function. For a vector $\mathbf{v}$ with non-negative entries we write $\mathrm{nrm}(\mathbf{v}) = \mathbf{v}/\mathbf{1}^\top\mathbf{v}$ if $\mathbf{1}^\top\mathbf{v} > 0$ and $\mathrm{nrm}(\mathbf{0}) = \mathbf{0}$, so that genre vectors live in $\bar{\Delta}^{\lvert\mathcal{G}\rvert} = \Delta^{\lvert\mathcal{G}\rvert} \cup \{\mathbf{0}\}$, the probability simplex extended by the "no evidence" point.

A. Seekers, sessions and the temporal boundary

ReDial is a set of dialogues, each a message sequence $D = (m_1, \ldots, m_T)$ in which every message carries a role (seeker or recommender), English text and the set $M(m) \subseteq \mathcal{I}$ of movies it mentions. Every dialogue has a seeker identifier u and an integer conversation identifier c, and the seeker annotates each mentioned movie as liked, seen and suggested. We write $\mathcal{L}(D) \subseteq \mathcal{I}$ for the movies the seeker marked as liked or seen in D.

Conversation identifiers were assigned sequentially during collection, and we treat them as the chronological order of a seeker's sessions (an assumption; see Section VIII). The history available at dialogue c of seeker u is then

$$\mathcal{H}_u(c) = \{D^u_{c'} : c' < c\}, \tag{1}$$

the set of u's dialogues with a strictly smaller identifier. Nothing from D^u_c itself or from any later dialogue enters the long-term side.

B. Instances

A recommendation instance is a triple (c, t, y): at turn $t > 1$ of dialogue c the recommender's message mentions a movie

$y \in \mathcal{I}$ that has not appeared earlier in the dialogue,

$$y \in M(m_t) \setminus \bigcup_{t'<t} M(m_{t'}), \qquad \mathrm{role}(m_t) = \text{recommender}. \tag{2}$$

A recommender message introducing several new movies yields one instance per movie. The prediction problem is to rank the full catalogue $\mathcal{I}$ so that y is placed as high as possible, using only the two evidence bundles below.

a) Long-term preference (LTP): From $\mathcal{H}_u(c)$ in chronological order we take the liked/seen items, keep the most recent $L_{\max} = 50$ of them, and call the result $\mathbf{h} = (h_1, \ldots, h_L)$ with h_L the most recent. We also keep the last ten seeker utterances in $\mathcal{H}_u(c)$ that match a preference-statement pattern ("I usually like…", "my favourite…") and are shorter than 200 characters; their concatenation is the profile text $\mathcal{P}$. The LTP genre vector is a recency-decayed mixture of the items' genre indicators,

$$\boldsymbol{\pi}_{\mathrm{L}} = \mathrm{nrm}\Big(\sum_{j=1}^{L} \gamma^{L-j} \frac{\mathbf{g}(h_j)}{|G(h_j)|}\Big), \qquad \mathbf{g}(i)_g = \mathbb{1}[g \in G(i)], \tag{3}$$

with $\gamma = 0.9$, so the most recent item has weight one and each older item is discounted by a further factor γ. Items with no MovieLens match ($G(i) = \emptyset$) contribute nothing. A seeker with $L < L_{\min} = 3$ history items is *cold*.

b) Short-term intent (STI): Let $S_{<t}$ be the seeker messages before turn t, R_t the concatenated text of the last 6 of them, and ℓ_t the text of the last one. Let $\mathcal{M}_{<t}$ be the movies mentioned in $S_{<t}$ and $\mathcal{M}^+_{<t} \subseteq \mathcal{M}_{<t}$ those the seeker annotated as liked. Titles in the text are substituted by their surface form, so "@12345" becomes the movie name. The STI genre vector combines lexicon counts with the genres of liked in-dialogue movies,

$$\boldsymbol{\pi}_{\mathrm{S}} = \mathrm{nrm}\Big(\mathbf{n}(R_t) + \mathrm{nrm}\Big(\sum_{i \in \mathcal{M}^+_{<t}} \frac{\mathbf{g}(i)}{|G(i)|}\Big)\Big), \tag{4}$$

where $\mathbf{n}(R_t)_g$ counts occurrences in R_t of the keywords of a small English lexicon for genre g ("scary", "slasher", "zombie" for Horror, and so on). Eight scalar flags summarise lexical and length evidence,

$$\mathbf{f} = \big(f_{\mathrm{sti}},\, f_{\mathrm{ltp}},\, f_{\mathrm{neg}},\, f_{\mathrm{req}},\, \tfrac{|S_{<t}|}{10},\, \tfrac{|\mathcal{M}_{<t}|}{5},\, f_{\mathrm{cold}},\, \tfrac{L}{L_{\max}}\big), \tag{5}$$

where f_{sti} and f_{ltp} indicate an explicit departure marker ("for a change", "tonight", "with my kids") or an explicit habit marker ("as usual", "stick with") in R_t, f_{neg} indicates a negated genre keyword ("not into horror"), f_{req} indicates a request pattern in ℓ_t, and $f_{\mathrm{cold}} = \mathbb{1}[L < L_{\min}]$.

Everything on the STI side is computed from $S_{<t}$ only; everything on the LTP side from $\mathcal{H}_u(c)$ only. The model therefore sees neither the target turn nor any later data.

C. Relationships, actions and shifts

We distinguish five relationship classes $\mathcal{R} = \{$Complement, Consistent, Conflict, Override, Uncertain$\}$ between the two bundles. *Consistent*: STI points where LTP already points. *Complement*: STI adds a genre that LTP does not cover without contradicting it. *Conflict*: STI opposes LTP with no sign that the user is aware of it. *Override*: STI opposes LTP and the user flags the departure explicitly. *Uncertain*: one side carries too little evidence. The action set is $\mathcal{A} = \{$Fuse, Prioritise_LTP, Prioritise_STI, Ask_Clarification$\}$, and a fixed default map $A_0 : \mathcal{R} \to \mathcal{A}$ sends Consistent and Complement to Fuse, Conflict and Override to Prioritise_STI, and Uncertain to Ask_Clarification. A learned policy may depart from A_0.

A departure from habit that occurs in one session is a *temporary override*. If the same departure wins arbitration in at least k distinct sessions of the same seeker it is a *persistent shift* and should be folded into LTP (Section IV-H).

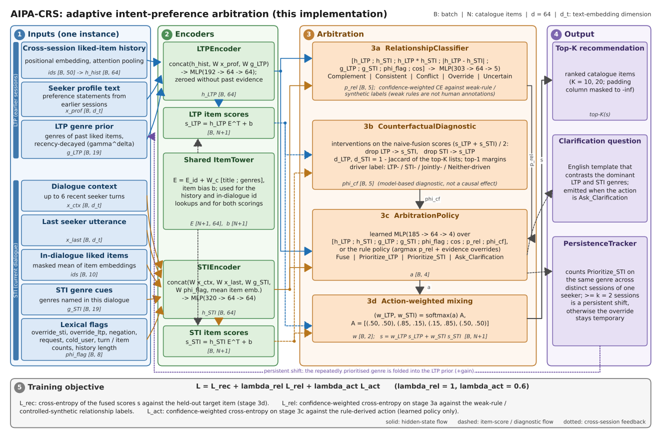

Fig. 1. AIPA components. History and dialogue context feed separate LTP and STI encoders. A relationship classifier and a counterfactual driver probe inform the arbitration policy, which produces fusion weights (or a clarifying question). A persistence tracker decides whether repeated intent shifts should update the long-term profile.

IV. Method

Figure 1 shows the components. All encoders are small because the point of the work is the arbitration logic, not representational power; the item tower, the dimensionality $d = 64$ and the training recipe are shared by every compared system so that differences are attributable to the arbitration design. Throughout, $\mathrm{MLP}(\mathbf{z}) = \mathbf{W}_2\,\mathrm{GELU}(\mathbf{W}_1\mathbf{z} + \mathbf{b}_1) + \mathbf{b}_2$ with hidden width d and dropout 0.1 after the activation; each occurrence has its own parameters.

A. Text and item representations

Text is encoded by a frozen map ϕ that is fixed before training. In the run reported here ϕ is the pre-trained sentence encoder all-MiniLM-L6-v2 [16], whose mean-pooled output is ℓ_2-normalised so that $d_t = 384$; the code also offers a lighter alternative fitted on the training split (word uni- and bigram TF–IDF with sublinear term frequency followed by truncated SVD),

$$\phi(x) = \frac{\psi(x)}{\|\psi(x)\|_2 + 10^{-8}}, \qquad \phi(\text{empty}) = \mathbf{0}, \tag{6}$$

with ψ either the MiniLM output or $\mathbf{V}^\top \mathrm{tfidf}(x)$. Embeddings are cached on disk by content hash, so a re-run encodes only new strings. Item i has a content vector $\mathbf{x}_i = [\phi(\text{title}_i); \mathbf{g}(i)] \in \mathbb{R}^{d_t+|\mathcal{G}|}$ and the item tower produces

$$\mathbf{e}_i = \mathbf{E}_i + \mathbf{W}_c\mathbf{x}_i + \mathbf{b}_c, \qquad b_i \in \mathbb{R}, \tag{7}$$

with a free embedding table $\mathbf{E} \in \mathbb{R}^{|\mathcal{I}|\times d}$, a shared projection $\mathbf{W}_c$ and a free per-item bias b_i. Given any query vector $\mathbf{h} \in \mathbb{R}^d$ the score of item i is the biased inner product

$$s_i(\mathbf{h}) = \mathbf{h}^\top \mathbf{e}_i + b_i, \tag{8}$$

and $\mathbf{s}(\mathbf{h}) \in \mathbb{R}^{|\mathcal{I}|}$ collects the scores of the whole catalogue (the padding index is masked to $-\infty$).

B. LTP encoder

The history $\mathbf{h}$ is right-aligned in a window of length $L_{\max}$ and each position j receives a learned positional vector $\mathbf{u}_j$. The encoder pools the item vectors with a single learned attention query $\mathbf{a} \in \mathbb{R}^d$,

$$\tilde{\mathbf{e}}_j = \mathbf{e}_{h_j} + \mathbf{u}_j, \qquad \alpha_j = \frac{\exp(\mathbf{a}^\top \tilde{\mathbf{e}}_j)}{\sum_{j'=1}^{L} \exp(\mathbf{a}^\top \tilde{\mathbf{e}}_{j'})}, \tag{9}$$

$$\mathbf{h}_{\text{hist}} = \sum_{j=1}^{L} \alpha_j \, \tilde{\mathbf{e}}_j, \tag{10}$$

where the softmax runs over the non-padding positions only and $\mathbf{h}_{\text{hist}} = \mathbf{0}$ when $L = 0$. The positional vectors let the attention prefer recent or old items, but the recency discount itself is applied to the genre vector through γ in Eq. (3) rather than to the attention weights. The encoding is

$$\mathbf{h}_{\text{L}} = \mathbb{1}[\text{LTP evidence}] \cdot \text{MLP}\big([\mathbf{h}_{\text{hist}};\, \mathbf{W}_p\phi(\mathcal{P});\, \mathbf{W}_g\boldsymbol{\pi}_{\text{L}}]\big), \tag{11}$$

where the indicator is one if at least one of $\mathbf{h}$, $\boldsymbol{\pi}_{\text{L}}$ or $\phi(\mathcal{P})$ is non-zero. A seeker without any history thus has $\mathbf{h}_{\text{L}} = \mathbf{0}$ and, by Eq. (8), an LTP score vector equal to the item biases.

C. STI encoder

The intent side has no sequence to attend over; it concatenates five projected views of the current dialogue, one of them a masked mean of the liked in-dialogue items (at most the last ten),

$$\bar{\mathbf{e}}_{\text{cur}} = \frac{1}{\max(1, |\mathcal{M}^+_{<t}|)} \sum_{i\in\mathcal{M}^+_{<t}} \mathbf{e}_i, \tag{12}$$

$$\mathbf{h}_{\text{S}} = \text{MLP}\big([\mathbf{W}_r\phi(R_t);\, \mathbf{W}_\ell\phi(\ell_t);\, \mathbf{W}'_g\boldsymbol{\pi}_{\text{S}};\, \mathbf{W}_f\mathbf{f};\, \bar{\mathbf{e}}_{\text{cur}}]\big). \tag{13}$$

D. Scoring and the fusion family

Both encodings are scored against the catalogue with Eq. (8), $\mathbf{s}_{\text{L}} = \mathbf{s}(\mathbf{h}_{\text{L}})$ and $\mathbf{s}_{\text{S}} = \mathbf{s}(\mathbf{h}_{\text{S}})$, and every compared system produces its final ranking from a convex combination

$$\mathbf{s} = w_{\text{L}}\, \mathbf{s}_{\text{L}} + w_{\text{S}}\, \mathbf{s}_{\text{S}}, \qquad w_{\text{L}} + w_{\text{S}} = 1,\ w_{\text{L}}, w_{\text{S}} \geq 0. \tag{14}$$

Because the score is linear in the query, Eq. (14) equals $\mathbf{s}(w_{\text{L}}\mathbf{h}_{\text{L}} + w_{\text{S}}\mathbf{h}_{\text{S}})$: mixing scores and mixing encodings are the same operation, and the item bias enters exactly once. The systems differ only in how $(w_{\text{L}}, w_{\text{S}})$ is chosen:

$$(w_{\text{L}}, w_{\text{S}}) = \begin{cases} (1, 0) & \text{LTP-only}, \\ (0, 1) & \text{STI-only}, \\ (\alpha, 1-\alpha) & \text{naive fusion},\ \alpha = 0.5, \\ (\sigma(z), 1-\sigma(z)) & \text{adaptive fusion}, \\ \mathbf{q}^\top \mathbf{W}_A & \text{AIPA}. \end{cases} \tag{15}$$

The naive weight is also swept over $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ at test time (Table IX). Adaptive fusion learns a scalar gate

$$z = \text{MLP}\big([\mathbf{h}_{\text{L}}; \mathbf{h}_{\text{S}}; \boldsymbol{\pi}_{\text{L}}; \boldsymbol{\pi}_{\text{S}}; \mathbf{f}]\big) \in \mathbb{R}, \tag{16}$$

and has no notion of relationship or action. AIPA replaces the gate by a distribution $\mathbf{q} \in \Delta^4$ over actions (Section IV-G) and a table that states what each action means for the weights. Each row of the table is the softmax of two free logits $\boldsymbol{\ell}_a \in \mathbb{R}^2$,

$$[\mathbf{W}_A]_{a\cdot} = \text{softmax}(\boldsymbol{\ell}_a), \qquad \mathbf{W}_A^{(0)} = \begin{array}{cl} \begin{pmatrix} 0.50 & 0.50 \\ 0.85 & 0.15 \\ 0.15 & 0.85 \\ 0.50 & 0.50 \end{pmatrix} & \begin{array}{l} \text{Fuse} \\ \text{Prioritise\_LTP} \\ \text{Prioritise\_STI} \\ \text{Ask\_Clarification} \end{array} \end{array} \tag{17}$$

so that $w_{\text{L}} = \sum_a q_a [\mathbf{W}_A]_{a1}$ and the constraint in Eq. (14) holds automatically. The logits are initialised so that $\mathbf{W}_A$ equals the hand-set table $\mathbf{W}_A^{(0)}$ and are then trained end-to-end through the recommendation loss (`learned_action_weights`); the previous fixed-table behaviour is retained as a configuration option. The meaning of each action therefore remains inspectable (one weight pair per action), but the pair itself is fitted; in the run reported here the Prioritise_STI row moved to roughly $0.22/0.78$ and the Fuse row stayed near $0.5/0.5$ (Table XI). An optional instance-level residual $g \in \mathbb{R}$ from the policy, $w_{\text{L}} \leftarrow \sigma(\text{logit}(w_{\text{L}}) + g)$, is implemented but switched off in every reported configuration. A clarifying turn still ranks the catalogue with the Ask_Clarification row, so an offline ranking metric is defined for every instance.

E. Relationship classifier

The classifier sees both encodings, their element-wise interactions, the raw genre vectors and flags, and the cosine of the encodings,

$$\boldsymbol{\varphi}_{\text{rel}} = \big[\mathbf{h}_{\text{L}};\, \mathbf{h}_{\text{S}};\, \mathbf{h}_{\text{L}} \odot \mathbf{h}_{\text{S}};\, |\mathbf{h}_{\text{L}} - \mathbf{h}_{\text{S}}|;\, \boldsymbol{\pi}_{\text{L}};\, \boldsymbol{\pi}_{\text{S}};\, \mathbf{f};\, \cos(\mathbf{h}_{\text{L}}, \mathbf{h}_{\text{S}})\big], \tag{18}$$

$$\mathbf{r} = \text{MLP}(\boldsymbol{\varphi}_{\text{rel}}) \in \mathbb{R}^5, \qquad \mathbf{p} = \text{softmax}(\mathbf{r}), \tag{19}$$

with $\cos(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top\mathbf{y} / \max(\|\mathbf{x}\|\|\mathbf{y}\|, 10^{-6})$. The posterior $\mathbf{p}$ is passed to the policy through a stop-gradient, so the action loss cannot reshape the relationship head; the relationship head is trained only by its own (weak) labels.

F. Counterfactual driver diagnostic

The diagnostic asks how much a ranking depends on each pathway, and answers by intervening on the model rather than on the data. The reference ranking is the naive fusion of the two component scores, and removing a pathway is defined as scoring with the other pathway alone,

$$\mathbf{s}_{\varnothing} = \tfrac{1}{2}(\mathbf{s}_{\mathrm{L}} + \mathbf{s}_{\mathrm{S}}), \quad \mathbf{s}_{\varnothing}\big|_{do(\mathrm{L}=\varnothing)} := \mathbf{s}_{\mathrm{S}}, \quad \mathbf{s}_{\varnothing}\big|_{do(\mathrm{S}=\varnothing)} := \mathbf{s}_{\mathrm{L}}. \tag{20}$$

This is a convention rather than a literal $\mathbf{h}_{\mathrm{L}} \leftarrow \mathbf{0}$ (which would give $\frac{1}{2}\mathbf{h}_{\mathrm{S}}^{\top}\mathbf{e}_i + b_i$ and weigh the item bias twice as heavily); it keeps the interventional list equal to what the STI or LTP tower would recommend on its own. Disruption is one minus the Jaccard overlap of the factual and interventional top-K lists ($K = 10$),

$$J_K(\mathbf{s}, \mathbf{s}') = \frac{|\mathrm{Top}_K(\mathbf{s}) \cap \mathrm{Top}_K(\mathbf{s}')|}{|\mathrm{Top}_K(\mathbf{s}) \cup \mathrm{Top}_K(\mathbf{s}')|}, \tag{21}$$

$$\delta_{\mathrm{L}} = 1 - J_K(\mathbf{s}_{\varnothing}, \mathbf{s}_{\mathrm{S}}), \qquad \delta_{\mathrm{S}} = 1 - J_K(\mathbf{s}_{\varnothing}, \mathbf{s}_{\mathrm{L}}), \tag{22}$$

so δ_{L} measures how much the list moves when LTP is removed and δ_{S} when STI is removed. Together with the top-1 margins $m(\mathbf{s}) = s_{(1)} - s_{(2)}$ of the three score vectors they form the feature vector handed to the policy,

$$\mathbf{c} = \big(\delta_{\mathrm{L}},\, \delta_{\mathrm{S}},\, m(\mathbf{s}_{\varnothing}),\, m(\mathbf{s}_{\mathrm{S}}),\, m(\mathbf{s}_{\mathrm{L}})\big) \in \mathbb{R}^5, \tag{23}$$

computed without gradient. The per-turn driver label uses a threshold $\tau = 0.1$ and a dominance ratio $\rho = 1.5$:

$$\text{driver} = \begin{cases} \text{Neither} & \delta_{\mathrm{L}} < \tau \ \wedge \ \delta_{\mathrm{S}} < \tau, \\ \text{STI} & \delta_{\mathrm{S}} \geq \tau \ \wedge \ \delta_{\mathrm{S}} \geq \rho\, \delta_{\mathrm{L}}, \\ \text{LTP} & \delta_{\mathrm{L}} \geq \tau \ \wedge \ \delta_{\mathrm{L}} \geq \rho\, \delta_{\mathrm{S}}, \\ \text{Jointly} & \text{otherwise}, \end{cases} \tag{24}$$

with the cases tested in the order written. The labels describe the sensitivity of the fitted model; nothing here identifies an effect on users. A second, post-training probe re-scores the *fused* model of Eq. (14) under $\mathbf{h}_{\mathrm{L}} \leftarrow \mathbf{0}$ and $\mathbf{h}_{\mathrm{S}} \leftarrow \mathbf{0}$ (which also changes $\mathbf{p}$, $\mathbf{q}$ and hence the weights) and reports rank and NDCG changes and the plain overlap $|\mathrm{Top}_{10} \cap \mathrm{Top}'_{10}|/10$; that is the quantity in Table VIII.

G. Arbitration policy and clarification trigger

a) Learned policy: The action logits are a function of the encodings, the evidence, the (detached) relationship posterior and the counterfactual features,

$$\boldsymbol{\varphi}_{\mathrm{act}} = \big[\mathbf{h}_{\mathrm{L}};\, \mathbf{h}_{\mathrm{S}};\, \boldsymbol{\pi}_{\mathrm{L}};\, \boldsymbol{\pi}_{\mathrm{S}};\, \mathbf{f};\, \cos(\mathbf{h}_{\mathrm{L}}, \mathbf{h}_{\mathrm{S}});\, \mathrm{sg}(\mathbf{p});\, \mathbf{c}\big], \tag{25}$$

$$\mathbf{o} = \mathrm{MLP}(\boldsymbol{\varphi}_{\mathrm{act}}) \in \mathbb{R}^4, \qquad \mathbf{q} = \mathrm{softmax}(\mathbf{o}), \tag{26}$$

where sg is the stop-gradient. The ablations remove $\mathrm{sg}(\mathbf{p})$ or $\mathbf{c}$ from Eq. (25); the "without clarification" ablation sets $o_{\mathrm{Ask}} = -\infty$ before the softmax. The weights follow from Eq. (17) and the emitted action is $\hat{a} = \arg\max_a q_a$.

b) Rule policy: The rule variant replaces Eq. (26) by a deterministic function of the relationship posterior and two evidence tests. With $\hat{r} = \arg\max_r p_r$, confidence $\kappa = \max_r p_r$, threshold $\theta = 0.5$, cold $= \mathbb{1}[f_{\mathrm{cold}} > \frac{1}{2}]$ and noSTI $= \mathbb{1}[\boldsymbol{\pi}_{\mathrm{S}} = \mathbf{0}]$,

$$\hat{a} = \begin{cases} \text{Prioritise\_STI} & \text{cold} \ \wedge \ \neg\text{noSTI}, \\ \text{Ask\_Clarification} & \kappa < \theta, \\ A_0(\hat{r}) & \text{otherwise}, \end{cases} \tag{27}$$

where the first case takes precedence. When clarification is disabled the second case is dropped, any Ask_Clarification produced by A_0 becomes Fuse, and a non-cold seeker with noSTI is sent to Prioritise_LTP. The rule output is turned into logits $\mathbf{o} = 20 \cdot \mathbf{1}_{\hat{a}}$, so its softmax is a near one-hot $\mathbf{q}$ and Eq. (17) returns the weight row of $\hat{a}$ almost exactly.

c) Clarification: A clarifying question is emitted whenever $\hat{a} =$ Ask_Clarification: for the learned policy this is the arg-max of Eq. (26), for the rule policy the second case of Eq. (27). The question is filled from a small English template set using the arg-max genres of $\boldsymbol{\pi}_{\mathrm{L}}$ and $\boldsymbol{\pi}_{\mathrm{S}}$, for example *"Earlier you enjoyed drama movies, but right now you seem interested in horror. Should I focus on horror for tonight, or find something that fits both?"* Offline we cannot observe the answer, so the question is evaluated by whether the reference label called for it (Section VI-B).

H. Persistence tracker

At inference the natural test instances are visited in order of (seeker, conversation id, turn). For each seeker u and genre g the tracker keeps the set $\mathcal{C}_{u,g}$ of sessions in which the first instance of the session was resolved by Prioritise_STI with g the arg-max of $\boldsymbol{\pi}_{\mathrm{S}}$,

$$\mathcal{C}_{u,g} \leftarrow \mathcal{C}_{u,g} \cup \{c\} \quad \text{if } \hat{a} = \text{Prioritise\_STI} \wedge g = \arg\max_{g'} \pi_{\mathrm{S},g'}. \tag{28}$$

Before an instance of seeker u is scored, its LTP genre vector is replaced by

$$\boldsymbol{\pi}'_{\mathrm{L}} = \mathrm{nrm}\Big(\boldsymbol{\pi}_{\mathrm{L}} + \eta \sum_{g:\, |\mathcal{C}_{u,g}| \geq k} \mathbf{1}_g\Big), \tag{29}$$

with $k = 2$ and $\eta = 0.3$, where $\mathbf{1}_g$ is the unit vector of genre g; only sessions already visited count, so the adjustment never uses the future. Each time $|\mathcal{C}_{u,g}|$ first reaches k a persistent shift (u, g, c) is logged. The model is then re-run with $\boldsymbol{\pi}'_{\mathrm{L}}$ in place of $\boldsymbol{\pi}_{\mathrm{L}}$; the encoders and the policy are unchanged, so the tracker acts only through Eqs. (11), (18), (16) and (25). Synthetic instances are excluded from the tracker.

I. Training objective

Every instance n carries a target y_n and, for AIPA variants, a relationship label $r_n \in \mathcal{R}$, an action label $a_n \in \mathcal{A}$ and a confidence $\omega_n \in [0, 1]$ from the labelling procedure of Section V ($\omega_n = 1$ for synthetic instances). With $\mathrm{CE}(\mathbf{z}, y) = -\log \mathrm{softmax}(\mathbf{z})_y$ and CE_ϵ its label-smoothed version, the mini-batch loss is

$$\mathcal{L} = \frac{1}{B} \sum_{n=1}^{B} \Big[\nu_n\, \mathcal{L}_{\mathrm{rec}}(\mathbf{s}_n, y_n) + \omega_n \big(\lambda_{\mathrm{rel}}\, \mathrm{CE}_\epsilon(\mathbf{r}_n, r_n) + \lambda_{\mathrm{act}}\, \mathrm{CE}(\mathbf{o}_n, a_n)\big)\Big], \tag{30}$$

with $\lambda_{\mathrm{rel}} = 1.0$, $\lambda_{\mathrm{act}} = 0.6$ and $\epsilon = 0.1$. Three ingredients target disagreement turns. *Conflict weighting.* $\nu_n = 2.0$ for instances whose label is Conflict or Override (natural weak-rule or synthetic) and 1 otherwise, renormalised to mean one within the batch so that the overall loss scale is unchanged (`conflict_loss_weight`; 1 recovers the unweighted loss). *Recommendation loss.* $\mathcal{L}_{\mathrm{rec}}$ is configurable as the full softmax over the catalogue, a sampled softmax with in-batch positives plus 256 uniform negatives, or BPR. The three were compared on the validation split of the development configuration (validation Hit@10 of AIPA (full): full softmax 0.123, sampled softmax 0.072, BPR 0.046) and the full softmax is used in every reported run. *Self-training.* From epoch 3 on, a weak-rule relationship label with $\omega_n < 0.6$ is replaced by the model's own prediction when its softmax maximum is at least 0.9; the decision is always taken against the original weak labels (so relabelling cannot drift), synthetic labels are never touched, and the action label follows A_0 of the new relationship. In the reported run this relabelled between 150 and 1,400 of the 36883 training instances per epoch (0.4–3.8%), the count growing as the head becomes more confident; training stopped early after 5–10 epochs on every seed. The two auxiliary terms are present only for AIPA variants, the action term only when the policy is learned, and each is scaled per instance by the label confidence so that a low-confidence weak label moves the heads less than a confident one. All of λ_{rel}, λ_{act}, the conflict weight, the smoothing and the self-training switch were selected on the validation split (natural validation Hit@10 as the primary criterion, validation conflict-subset Hit@10 and macro-F1 reported alongside; grid $\lambda_{\text{rel}} \in \{0.5, 1\} \times \lambda_{\text{act}} \in \{0.1, 0.6\}$, all within 0.002 Hit@10 of each other); the test split was not used for any selection. For the "without clarification" ablation the action label Ask_Clarification is remapped to Fuse in Eq. (30) as well as being masked in Eq. (26), so the ablation is not merely a decoding constraint. All models are trained with AdamW [17] (learning rate 0.003, weight decay 10^{-5}), batch size 256, gradient-norm clipping at 5, for at most 20 epochs with early stopping on validation Hit@10 (patience three) and restoration of the best epoch. Baselines that are not fusion models (GRU, conversation-aware) use the same item tower, Eq. (8) and the first term of Eq. (30).

V. Data and Labels

A. ReDial and cross-session histories

ReDial [7] contains 11,348 English dialogues (10,006 training, 1,342 test) between a seeker and a recommender, with per-movie seeker annotations for *liked*, *seen* and *suggested*. We split 1,000 training dialogues off as validation. The corpus has 764 distinct seeker identifiers and most seekers appear in many dialogues (Fig. 2). We order a seeker's dialogues by conversation identifier and treat this as chronological order; this is an assumption we could not verify against timestamps, and we flag it as such. For dialogue D_{m+1}, LTP is built only from $D_1, \ldots, D_m$; a seeker with fewer than 3 earlier sessions is treated as cold. Item metadata (genres, years) come from MovieLens *ml-latest* [18] joined on normalised title and year; movies without a match keep an empty genre vector rather than an imputed one.

Each recommendation instance is one (dialogue, turn, target movie) triple where the recommender introduces a movie not yet mentioned in the dialogue. Only the preceding 6 turns of context and at most 50 history items are used. In the run reported here we sample 100% of the dialogues in every split with a fixed seed, which yields 36883 training instances and 4472 natural test instances.

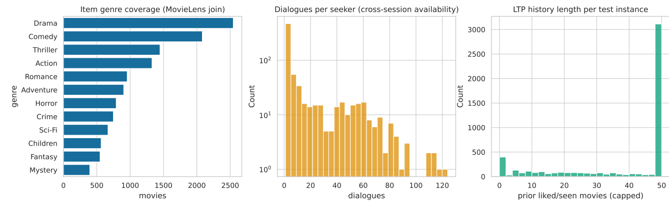

Fig. 2. ReDial overview: dialogue counts per split, dialogues per seeker, genre coverage after the MovieLens join, and turns per dialogue.

B. Relationship labels

ReDial has no relationship annotations. We use two label sources and keep them apart throughout.

Weak-rule labels. A deterministic function of $(\boldsymbol{\pi}_{\text{L}}, \boldsymbol{\pi}_{\text{S}}, R_t, \mathbf{f})$ assigns a relationship r_n, an action a_n and a confidence ω_n. Let $\text{sim} = \cos(\boldsymbol{\pi}_{\text{L}}, \boldsymbol{\pi}_{\text{S}})$, let $T(\boldsymbol{\pi})$ be the (at most two) largest genres of $\boldsymbol{\pi}$ with mass at least 0.15, and let $\text{hasL} = \mathbb{1}[\boldsymbol{\pi}_{\text{L}} \neq \mathbf{0} \wedge \neg f_{\text{cold}}]$. Three lexical tests read R_t: a habit marker (f_{ltp}), a departure marker (f_{sti}), and $\text{negH} = \mathbb{1}[\exists g \text{ negated in } R_t : \pi_{\text{L},g} \geq 0.2]$. Two structural tests compare the genre vectors: $\text{tens} = \mathbb{1}[\exists s \in T(\boldsymbol{\pi}_{\text{S}}),\, g \in T(\boldsymbol{\pi}_{\text{L}}) : \{s, g\} \in \mathcal{T} \wedge \pi_{\text{L},s} < 0.1]$ for a fixed list $\mathcal{T}$ of genre pairs that rarely co-occur in a request (Horror–Children, Comedy–War, ...), and $\text{new} = \mathbb{1}[\exists s \in T(\boldsymbol{\pi}_{\text{S}}) : \pi_{\text{L},s} < 0.05]$. Writing C/O for Override if $f_{\text{sti}} = 1$ and Conflict otherwise, the rule is evaluated top to bottom:

$$(r_n, a_n, \omega_n) = \begin{cases} (\text{Uncertain, Ask, } 0.9) & \boldsymbol{\pi}_{\text{S}} = \mathbf{0} \wedge \neg\text{hasL}, \\ (\text{Uncertain, Ask, } 0.6) & \boldsymbol{\pi}_{\text{S}} = \mathbf{0}, \\ (\text{Uncertain, Prio.\_STI, } 0.6) & \neg\text{hasL}, \\ (\text{Consistent, Fuse, } 0.8) & f_{\text{ltp}} = 1, \\ (\text{C/O, Prio.\_STI, } 0.8) & \text{negH}, \\ (\text{C/O, Prio.\_STI, } \omega_{\text{t}}) & \text{tens} \wedge \text{sim} < 0.25, \\ (\text{Consistent, Fuse, } \omega_{\text{s}}) & \text{sim} \geq 0.5, \\ (\text{Complement, Fuse, } \omega_{\text{n}}) & \text{new}, \\ (\text{Consistent, Fuse, } 0.5) & \text{sim} \geq 0.25, \\ (\text{Uncertain, Ask, } 0.5) & \text{otherwise}, \end{cases} \tag{31}$$

with $\omega_{\text{s}} = 0.5 + 0.5\,\text{sim}$, $\omega_{\text{n}} = 0.5 + 0.3(1 - \text{sim})$, $\omega_{\text{t}} = 0.75$ if $f_{\text{sti}} = 1$ and $\omega_{\text{t}} = 0.55 + 0.2\,(0.25 - \text{sim})/0.25$ otherwise. The action label follows A_0 except in the third case, where a cold seeker with a stated intent is labelled Prioritise_STI rather than Ask. These labels are noisy by construction; ω_n down-weights them in Eq. (30) and defines the "necessary clarification" event in Eq. (37).

Synthetic controlled labels. For 0.15 of natural test instances whose seeker has a usable LTP, we append one English seeker utterance and set the intended relationship. For *Conflict*, *Override* and *Consistent* the utterance requests a genre that conflicts with, overrides, or agrees with the seeker's top habitual genre, at one of three intensities (from a mild "maybe" to an insistent request), and the target is replaced by a movie of the requested genre. For *Complement* the requested genre is one that co-occurs with the habitual genres in the catalogue and is not in the tension list $\mathcal{T}$, so it adds a dimension without contradicting the profile. For *Uncertain* the utterance is deliberately vague ("anything is fine, surprise me"), all current-intent cues are removed and the target is left unchanged. Each synthetic instance records its source

TABLE II
RELATIONSHIP LABEL COUNTS BY SOURCE AND SPLIT.

Source	Relationship	Train	Test
weak_rule	Complement	5530	744
weak_rule	Conflict	3211	292
weak_rule	Consistent	16193	2070
weak_rule	Override	237	39
weak_rule	Uncertain	7178	1327
synthetic_controlled	Complement	996	134
synthetic_controlled	Conflict	742	112
synthetic_controlled	Consistent	1038	123
synthetic_controlled	Override	776	114
synthetic_controlled	Uncertain	982	141

instance, the injected utterance, the intended relationship and the intensity, and carries an explicit synthetic flag. Synthetic instances appear in the training set as well (with the same flag) so that the relationship head sees clean examples of the rarer classes.

Table II gives the resulting distribution. *Override* is rare in natural data, which is expected: people seldom announce that they are departing from their habits. A slot for human-verified labels exists in the pipeline; no such annotations were available for this run and every table that would have used them reports "not run".

VI. EXPERIMENTAL SETUP

A. Compared systems

LTP-only and *STI-only* score items from one encoder. *Naive fusion* averages the two encodings with $\alpha = 0.5$; *Adaptive fusion* learns a scalar gate from the two encodings but has no relationship or action head. *Sequential (GRU)* runs a GRU [12] over the concatenation of history items and current-dialogue mentions. *SASRec* [13] stacks two causal self-attention blocks over the same item sequence and scores from the last real position; it does not read dialogue text and is the strongest history-only comparison. *Conversation-aware* attends over the current dialogue with the history vector as query. *KBRD-style* follows the entity-centric design of KBRD [5]: items and genre "entities" mentioned in the dialogue are embedded and pooled with self-attention, fused through a learned gate with the dialogue text encoding, and scored with a genre-aware bias; the external knowledge graph and the response generator of the original are replaced by the shared item tower and MovieLens genres. All four are our own re-implementations in the same parameter budget and item tower; they are not reproductions of any published system and their numbers are not comparable to published ones. The AIPA ablations remove the relationship head, the counterfactual features, clarification, or the persistence tracker, or swap the learned policy for the rule table.

B. Metrics and statistics

a) Ranking: For an instance with target y and fused scores $\mathbf{s}$ the rank of the target is the number of catalogue items strictly ahead of it plus one, and the three ranking metrics are functions of that rank alone,

$$r = 1 + \left|\{i \in \mathcal{I} : s_i > s_y\}\right|, \tag{32}$$

$$\mathrm{Hit@}K = \mathbb{1}[r \le K], \qquad \mathrm{NDCG@}K = \frac{\mathbb{1}[r \le K]}{\log_2(r+1)},$$
$$\mathrm{MRR@}K = \frac{\mathbb{1}[r \le K]}{r}, \tag{33}$$

for $K \in \{10, 20\}$. With one target per instance Hit@K coincides with Recall@K. Reported values are means over test instances and over the 5 seeds.

b) Conflict subsets: Two natural conflict subsets are reported. The *strict* subset contains the natural test instances whose weak-rule label is Conflict or Override ($n = 331$). The *broad* subset adds instances whose weak label has confidence $\omega_n \ge 0.6$ and whose genre vectors disagree, $\mathrm{JS}(\boldsymbol{\pi}_{\mathrm{L}}, \boldsymbol{\pi}_{\mathrm{S}}) \ge 0.5$ (Jensen–Shannon divergence, base 2), giving $n = 1272$. Synthetic Conflict/Override injections form a third subset ($n = 226$) that is never pooled with the natural ones.

c) Relationship classification: With $\hat{r}_n = \arg\max_r p_{n,r}$ we report accuracy, per-class F1 and

$$\text{macro-F1} = \frac{1}{|\mathcal{R}|}\sum_{r\in\mathcal{R}} \mathrm{F1}_r, \qquad \text{weighted-F1} = \sum_{r\in\mathcal{R}} \frac{N_r}{N}\,\mathrm{F1}_r, \tag{34}$$

where $\mathrm{F1}_r$ is set to zero for a class that is never predicted and never occurs. Calibration uses the confidence $\kappa_n = \max_r p_{n,r}$ and ten equal-width bins $B_b = \{n : \kappa_n \in ((b-1)/10,\, b/10]\}$,

$$\mathrm{ECE} = \sum_{b=1}^{10} \frac{|B_b|}{N} \left| \frac{1}{|B_b|} \sum_{n\in B_b} \mathbb{1}[\hat{r}_n = r_n] - \frac{1}{|B_b|} \sum_{n\in B_b} \kappa_n \right|, \tag{35}$$

$$\mathrm{Brier} = \frac{1}{|\mathcal{R}^+|} \sum_{r\in\mathcal{R}^+} \frac{1}{N} \sum_{n=1}^{N} \left(p_{n,r} - \mathbb{1}[r_n = r]\right)^2, \tag{36}$$

where empty bins are skipped and $\mathcal{R}^+$ is the set of classes that occur at least once in the evaluated subset (one-versus-rest Brier score averaged over those classes) [19].

d) Arbitration and clarification: Let $\hat{a}_n$ be the emitted action, a_n the reference action, and define the per-instance events

$$\begin{aligned} \mathrm{asked}_n &= \mathbb{1}[\hat{a}_n = \text{Ask\_Clarification}], \\ \mathrm{conf}_n &= \mathbb{1}[r_n \in \{\text{Conflict}, \text{Override}\}], \\ \mathrm{nec}_n &= \mathbb{1}[r_n = \text{Uncertain}] \;\vee\; \left(\mathrm{conf}_n \wedge \omega_n < \theta_{\mathrm{c}}\right), \end{aligned} \tag{37}$$

with $\theta_{\mathrm{c}} = 0.6$: a question is deemed necessary when the reference relationship is Uncertain or when it is a conflict that the labelling rule itself was not confident about. Then

$$\begin{aligned} \text{clar. rate} &= \tfrac{1}{N}\textstyle\sum_n \mathrm{asked}_n, \\ \text{clar. precision} &= \frac{\sum_n \mathrm{asked}_n\, \mathrm{nec}_n}{\sum_n \mathrm{asked}_n}, \\ \text{clar. efficiency} &= \frac{\sum_n \mathrm{asked}_n\, \mathrm{nec}_n}{\sum_n \mathrm{nec}_n}, \\ \text{unnecessary rate} &= \tfrac{1}{N}\textstyle\sum_n \mathrm{asked}_n(1 - \mathrm{nec}_n), \end{aligned} \tag{38}$$

and the wrong-override rate is $\frac{1}{N}\sum_n \mathbb{1}[\hat{a}_n = \text{Prioritise\_STI} \wedge r_n = \text{Consistent}]$. Arbitration accuracy is $\frac{1}{N}\sum_n \mathbb{1}[\hat{a}_n = a_n]$; conflict-resolution accuracy is the same quantity restricted to $\text{conf}_n = 1$; override success is the mean Hit@10 over instances with $r_n = \text{Override}$ and $\hat{a}_n = \text{Prioritise\_STI}$. Precision is undefined when no question is asked.

e) Uncertainty and paired comparisons: Each metric x_n is first averaged over seeds for every test instance n. A 95% interval for a mean is the percentile bootstrap over instances with $B = 1000$ resamples,

$$\bar{x}^{(b)} = \frac{1}{N}\sum_{n=1}^{N} x_{\xi_n^{(b)}}, \quad \xi_n^{(b)} \overset{\text{iid}}{\sim} \text{Unif}\{1,\dots,N\}, \tag{39}$$

and the interval is $[\bar{x}^{(\cdot)}_{2.5\%}, \bar{x}^{(\cdot)}_{97.5\%}]$, the empirical percentiles of the resampled means, with a fixed resampling seed. AIPA (full) is compared with every other system on the same instances through the paired differences $d_n = x_n^{\text{AIPA}} - x_n^{\text{other}}$ using the paired t-test and the Wilcoxon signed-rank test (zero differences split between the signs). The m p-values of one family (one metric, one subset) are adjusted by the Holm step-down procedure [20]: with $p_{(1)} \le \cdots \le p_{(m)}$,

$$\tilde{p}_{(j)} = \max_{l \le j} \min\big(1, (m - l + 1)\, p_{(l)}\big). \tag{40}$$

Effect sizes are Cohen's d on the paired differences and Cliff's δ [21] over all instance pairs,

$$d = \frac{\bar{d}}{s_d}, \qquad \delta = \frac{1}{N^2}\sum_{i,j=1}^{N}\big(\mathbb{1}[x_i > x'_j] - \mathbb{1}[x_i < x'_j]\big), \tag{41}$$

where s_d is the sample standard deviation of the d_n, $x = x^{\text{AIPA}}$ and $x' = x^{\text{other}}$, and for $N > 2000$ the sum over j runs over the first 2000 control instances. A comparison whose differences are identically zero (e.g. an ablation that changed no prediction) is reported as undefined rather than as $p = 1$. All natural-data and synthetic-data results are reported separately.

C. Compute

Everything runs on CPU (8 cores; PyTorch 2.14.0+cpu, Python 3.12.8). The configuration used here ("full" mode) uses the whole corpus (100%), 5 seeds, 1000 bootstrap resamples, at most 20 epochs with early stopping on validation Hit@10, hidden size 64 and the MiniLM text encoder. Every hyper-parameter, including the loss weights $\lambda_{\text{rel}} = 1.0$, $\lambda_{\text{act}} = 0.6$, the conflict weight 2.0 and label smoothing 0.1, was chosen on the validation split of a smaller "quick" configuration before this run; the test split was scored once. The complete run takes about 72 minutes. Where we say "the earlier reduced-budget run" we mean the development configuration (25% of the corpus, two seeds, TF–IDF/SVD text), quoted only to show what changed with scale and encoder.

VII. Results

A. Overall ranking quality

Table III and Fig. 3 report the natural test set. Two things stand out. First, the intent signal dominates on ReDial: STI-only (0.125) more than doubles LTP-only (0.055), the two history-only sequential baselines (GRU 0.085, SASRec 0.079) add little over the profile, and every model that reads the dialogue lands between 0.116 and 0.128 Hit@10. Second, AIPA (full) at 0.117 sits at the lower end of that band. It is far better than LTP-only ($\Delta = +0.062$, Holm-corrected $p < 0.001$, $d = 0.19$), SASRec ($+0.038$, $p < 0.001$) and the GRU baseline ($+0.032$, $p < 0.001$); it is indistinguishable from naive fusion, adaptive fusion and its own rule-policy variant (all within 0.003, $p > 0.6$); and it is slightly but significantly *worse* than the conversation-aware baseline (-0.006, $p = 0.026$), STI-only ($\Delta = -0.009$, $p < 0.001$, $d = -0.03$) and the KBRD-style baseline (-0.011, $p < 0.001$, $d = -0.04$; Table IV). The effect sizes are tiny, but with 4,472 test turns and five seeds they are real. The pre-registered success criterion H1 (beat the best baseline on natural Hit@10) is therefore not met; the best system on this corpus is the KBRD-style entity-item model, with STI-only a close second.

TABLE III
NATURAL TEST SET ($n = 4472$, MEAN OVER 5 SEEDS; BRACKETS ARE 95% BOOTSTRAP INTERVALS). ROWS ABOVE THE RULE ARE BASELINES.

Model	Hit@10 [95% CI]	NDCG@10	MRR@10	Hit@20	NDCG@20
LTP-only	0.055 [0.052, 0.057]	0.030	0.023	0.084	0.037
STI-only	0.125 [0.121, 0.130]	0.066	0.048	0.196	0.084
Naive fusion	0.119 [0.115, 0.124]	0.063	0.046	0.191	0.081
Adaptive fusion	0.120 [0.116, 0.124]	0.062	0.045	0.187	0.079
Sequential (GRU)	0.085 [0.081, 0.088]	0.044	0.032	0.132	0.056
Conversation-aware	0.123 [0.118, 0.127]	0.065	0.047	0.189	0.081
SASRec	0.079 [0.075, 0.082]	0.038	0.026	0.125	0.050
KBRD-style	0.128 [0.123, 0.132]	0.067	0.048	0.195	0.083
AIPA w/o relationship	0.117 [0.113, 0.121]	0.061	0.044	0.187	0.078
AIPA w/o counterfactual	0.116 [0.112, 0.120]	0.060	0.044	0.186	0.078
AIPA w/o clarification	0.118 [0.114, 0.122]	0.062	0.045	0.184	0.079
AIPA w/o persistence	0.116 [0.113, 0.120]	0.060	0.044	0.181	0.076
AIPA (rule policy)	0.120 [0.115, 0.124]	0.062	0.045	0.187	0.079
AIPA (full)	**0.117 [0.113, 0.121]**	**0.060**	**0.043**	**0.180**	**0.076**

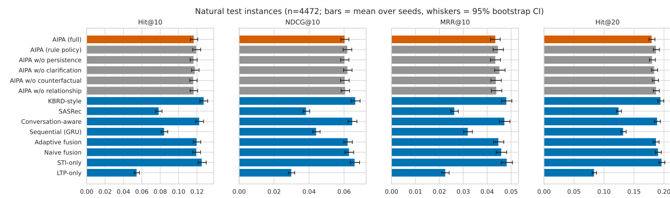

Fig. 3. Hit@10 and NDCG@10 on the natural test set with bootstrap intervals.

B. Conflict turns

The hypothesis that motivates the work is about the turns where LTP and STI disagree. Table V shows the strict natural conflict subset ($n = 331$ under weak-rule labels), the broad subset ($n = 1272$) and the controlled synthetic conflict subset ($n = 226$), and Fig. 4 contrasts conflict with non-conflict turns.

On strict natural conflicts the ordering is the same as on the whole set: KBRD-style 0.172, STI-only 0.169, adaptive fusion and the conversation-aware baseline 0.162, AIPA (full) 0.150, naive fusion 0.148, LTP-only 0.035. With 331 turns and five seeds the intervals still overlap, and AIPA's deficit to the top two (-0.019 to -0.022) is not significant after correction ($p_{\text{Holm}} = 0.16$), so on the strict subset the honest reading is "no evidence of a difference among the intent-reading models, point estimate against AIPA". The broad subset (1272 turns) removes the ambiguity: STI-only 0.164 and KBRD-style 0.163

TABLE IV
PAIRED COMPARISONS OF AIPA (FULL) AGAINST EACH OTHER SYSTEM ON HIT@10. Δ IS THE MEAN PER-INSTANCE DIFFERENCE (POSITIVE FAVOURS AIPA), p_{Holm} THE HOLM-CORRECTED PAIRED t-TEST p-VALUE (*: < 0.05), d COHEN'S d. DASHES MARK COMPARISONS WITH ZERO VARIANCE.

	Natural (all, n=22360)			Natural conflict (n=1655)			Synthetic conflict (n=1130)		
Control	Δ	p_{Holm}	d	Δ	p_{Holm}	d	Δ	p_{Holm}	d
LTP-only	+0.062	<0.001*	+0.19	+0.115	<0.001*	+0.31	+0.219	<0.001*	+0.51
STI-only	-0.009	<0.001*	-0.03	-0.019	0.161	-0.06	+0.009	1.000	+0.02
Naive fusion	-0.003	0.771	-0.01	+0.002	1.000	+0.01	+0.010	1.000	+0.03
Adaptive fusion	-0.003	0.644	-0.01	-0.012	0.998	-0.04	-0.018	1.000	-0.04
Sequential (GRU)	+0.032	<0.001*	+0.10	+0.065	<0.001*	+0.18	+0.230	<0.001*	+0.53
Conversation-aware	-0.006	0.026*	-0.02	-0.012	1.000	-0.03	+0.019	1.000	+0.05
SASRec	+0.038	<0.001*	+0.11	+0.079	<0.001*	+0.22	+0.240	<0.001*	+0.55
KBRD-style	-0.011	<0.001*	-0.04	-0.022	0.161	-0.06	-0.012	1.000	-0.03
AIPA w/o relationship	+0.000	1.000	+0.00	-0.003	1.000	-0.01	+0.004	1.000	+0.01
AIPA w/o counterfactual	+0.001	1.000	+0.00	-0.003	1.000	-0.01	+0.004	1.000	+0.01
AIPA w/o clarification	-0.001	1.000	-0.01	+0.003	1.000	+0.01	+0.004	1.000	+0.01
AIPA w/o persistence	+0.000	1.000	+0.00	+0.005	0.232	+0.05	–	–	–
AIPA (rule policy)	-0.003	0.771	-0.01	+0.006	1.000	+0.02	+0.014	1.000	+0.04

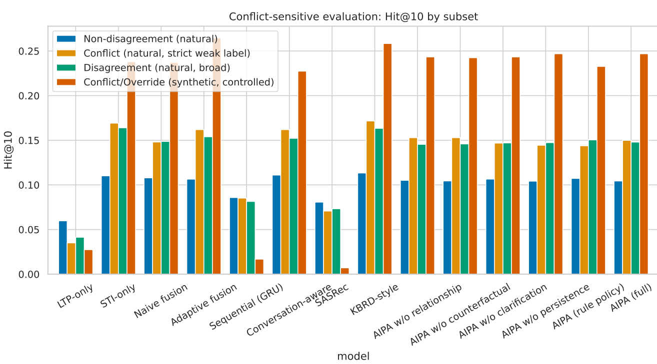

Fig. 4. Hit@10 on natural conflict versus non-conflict turns. Intervals on the conflict side are wide because the subset is small.

Fig. 5. Hit@10 per relationship class on natural (weak-rule) and synthetic labels.

are significantly ahead of AIPA (full) at 0.148 ($p_{\mathrm{Holm}} < 0.02$), while adaptive fusion (0.154) and naive fusion (0.149) are not distinguishable from it. On synthetic conflicts every model that trusts STI does well (0.23–0.27), LTP-only, GRU and SASRec collapse to below 0.03, and AIPA (full) at 0.247 sits between adaptive fusion (0.265, the best) and STI-only (0.238) with no significant difference to either. The synthetic set therefore confirms that AIPA learns to follow a clearly stated departure, but it also shows that "always follow the intent" is an equally good policy on this data. Criteria H2 (strict and broad) and H3 are not met.

Figure 5 breaks the natural set down by all five relationship classes. The *Uncertain* class (1,327 instances, 30% of them cold seekers) is where the KBRD-style and conversation-aware baselines gain most of their edge (0.104 and 0.103 versus 0.096 for AIPA (full) and STI-only); on *Consistent* turns, the majority, all fused models are within a few thousandths of each other. Split by history length (Fig. 10), AIPA (full) is competitive on cold seekers (0.133 versus 0.129 for STI-only and 0.138 for KBRD-style) and on 10–24 history items (0.124 versus 0.128 and 0.131), and falls behind on the 3,273 long-history turns (0.116 versus 0.126 and 0.128), where a profile that is trusted too much appears to cost more than it adds.

C. Relationship classification and arbitration

Table VI reports the relationship head. On natural data accuracy is 0.80 and macro-F1 0.62–0.66 (AIPA (full): 0.627, which meets the pre-set criterion of 0.5 and is up from 0.54 in the earlier reduced-budget run); the head is good at *Consistent* and *Uncertain* (F1 0.85 and 0.95), fair at *Complement* (0.55) and *Conflict* (0.41, up from below 0.2 but still short of the 0.5 criterion), and weakest at *Override* (0.37), which has only 39 natural test instances. Two caveats: the reference labels are themselves weak, so part of this "error" is label noise, and the two classes the head struggles with are exactly the two the rule defines through a similarity threshold. On synthetic data the head is near-perfect (macro-F1 0.96; *Conflict* 0.93, *Override* 0.99, injected vague *Uncertain* utterances 1.00, injected *Complement* requests 0.93), which shows it recognises the patterns when the text states them plainly. The confusion matrix in Fig. 6 shows that the natural conflicts it misses are mostly absorbed into *Consistent*.

The arbitration head (Table VII, Fig. 7) does what it was designed to do. The learned policy asks a clarifying question on about 20% of natural turns, and when it asks, the reference label is *Uncertain* or a low-confidence conflict 99% of the time; unnecessary clarifications are 0.2% and wrong overrides 1%, so both arbitration criteria are met. The rule policy asks far more often (32%) with markedly lower precision (0.64)

TABLE V
CONFLICT SUBSETS, HIT@10 AND COMPANIONS. NATURAL CONFLICTS USE WEAK-RULE LABELS (STRICT: CONFLICT/OVERRIDE; BROAD: PLUS CONFIDENT GENRE DISAGREEMENT); SYNTHETIC CONFLICTS ARE CONTROLLED INJECTIONS. SAME LAYOUT AS TABLE III.

Natural conflict, strict (n = 331)

Model	Hit@10 [95% CI]	NDCG@10	MRR@10	Hit@20	NDCG@20
LTP-only	0.035 [0.027, 0.044]	0.018	0.013	0.057	0.024
STI-only	0.169 [0.152, 0.187]	0.088	0.064	0.260	0.111
Naive fusion	0.148 [0.132, 0.165]	0.077	0.056	0.246	0.102
Adaptive fusion	0.162 [0.144, 0.180]	0.082	0.058	0.253	0.105
Sequential (GRU)	0.085 [0.071, 0.099]	0.044	0.032	0.132	0.056
Conversation-aware	0.162 [0.143, 0.179]	0.081	0.057	0.240	0.101
SASRec	0.071 [0.059, 0.083]	0.033	0.022	0.110	0.043
KBRD-style	0.172 [0.152, 0.190]	0.091	0.067	0.265	0.115
AIPA w/o relationship	0.153 [0.136, 0.169]	0.075	0.051	0.236	0.096
AIPA w/o counterfactual	0.153 [0.137, 0.169]	0.075	0.052	0.239	0.097
AIPA w/o clarification	0.147 [0.130, 0.163]	0.074	0.053	0.233	0.096
AIPA w/o persistence	0.144 [0.129, 0.161]	0.073	0.051	0.234	0.095
AIPA (rule policy)	0.144 [0.127, 0.160]	0.070	0.048	0.225	0.090
AIPA (full)	**0.150 [0.134, 0.166]**	**0.075**	**0.053**	**0.232**	**0.095**

Natural conflict, broad (n = 1272)

Model	Hit@10 [95% CI]	NDCG@10	MRR@10	Hit@20	NDCG@20
LTP-only	0.041 [0.037, 0.046]	0.020	0.014	0.067	0.027
STI-only	0.164 [0.155, 0.173]	0.083	0.059	0.251	0.105
Naive fusion	0.149 [0.139, 0.157]	0.076	0.054	0.237	0.098
Adaptive fusion	0.154 [0.145, 0.162]	0.076	0.053	0.236	0.097
Sequential (GRU)	0.081 [0.075, 0.089]	0.041	0.029	0.130	0.054
Conversation-aware	0.152 [0.144, 0.161]	0.077	0.055	0.228	0.097
SASRec	0.073 [0.067, 0.080]	0.035	0.024	0.117	0.046
KBRD-style	0.163 [0.154, 0.172]	0.085	0.061	0.247	0.105
AIPA w/o relationship	0.145 [0.137, 0.154]	0.073	0.051	0.229	0.094
AIPA w/o counterfactual	0.146 [0.138, 0.154]	0.074	0.053	0.228	0.095
AIPA w/o clarification	0.147 [0.138, 0.155]	0.075	0.053	0.226	0.095
AIPA w/o persistence	0.147 [0.139, 0.156]	0.073	0.051	0.224	0.093
AIPA (rule policy)	0.150 [0.142, 0.160]	0.075	0.053	0.232	0.096
AIPA (full)	**0.148 [0.140, 0.156]**	**0.073**	**0.051**	**0.223**	**0.092**

Synthetic conflict (n = 226)

Model	Hit@10 [95% CI]	NDCG@10	MRR@10	Hit@20	NDCG@20
LTP-only	0.027 [0.019, 0.036]	0.010	0.005	0.055	0.017
STI-only	0.238 [0.215, 0.263]	0.124	0.090	0.377	0.160
Naive fusion	0.237 [0.213, 0.263]	0.122	0.087	0.382	0.158
Adaptive fusion	0.265 [0.239, 0.292]	0.135	0.095	0.404	0.170
Sequential (GRU)	0.017 [0.010, 0.025]	0.006	0.003	0.035	0.010
Conversation-aware	0.227 [0.204, 0.252]	0.124	0.093	0.343	0.153
SASRec	0.007 [0.003, 0.012]	0.002	0.001	0.016	0.005
KBRD-style	0.258 [0.233, 0.283]	0.140	0.104	0.384	0.172
AIPA w/o relationship	0.243 [0.219, 0.269]	0.128	0.093	0.381	0.162
AIPA w/o counterfactual	0.242 [0.217, 0.269]	0.124	0.088	0.352	0.151
AIPA w/o clarification	0.243 [0.219, 0.269]	0.123	0.088	0.394	0.161
AIPA w/o persistence	0.247 [0.219, 0.272]	0.123	0.086	0.382	0.157
AIPA (rule policy)	0.233 [0.209, 0.258]	0.125	0.092	0.379	0.161
AIPA (full)	**0.247 [0.219, 0.272]**	**0.123**	**0.086**	**0.382**	**0.157**

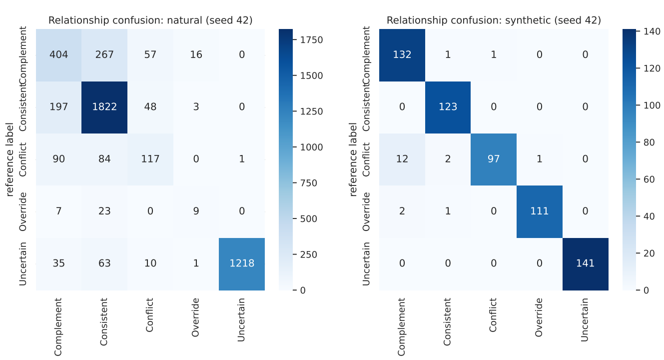

Fig. 6. Relationship confusion matrix of AIPA (full) on natural test turns (rows: weak-rule reference; columns: prediction).

and an unnecessary-clarification rate of 12%, so the learned policy is doing more than reproducing the table it was seeded with. Conflict-resolution accuracy on natural conflicts is low (0.35 for AIPA (full), 0.29 for the rule) for the reasons above; on synthetic conflicts it is 0.94. Calibration of the relationship head is good (ECE 0.03–0.06 on natural data, Fig. 8).

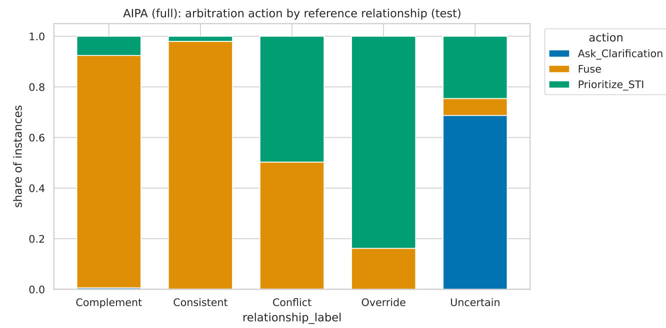

Fig. 7. Distribution of predicted actions within each reference relationship class, AIPA (full), natural turns.

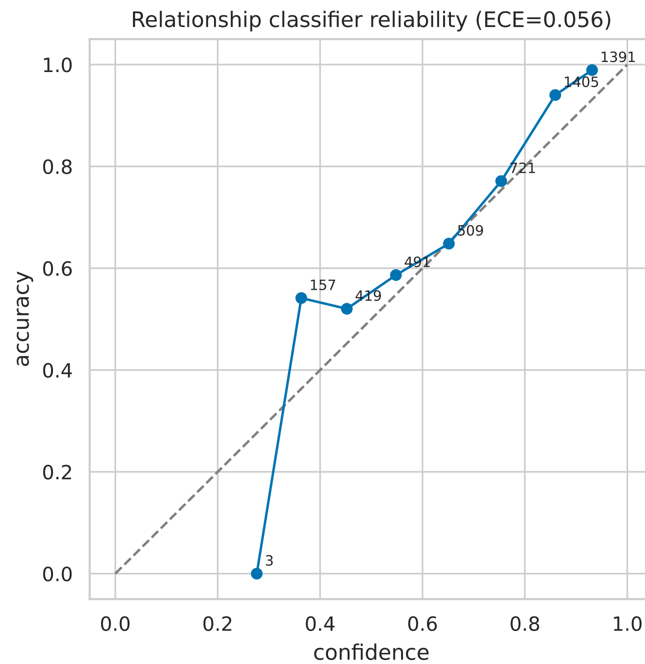

Fig. 8. Reliability diagram of the relationship head.

D. Counterfactual driver diagnostic

Table VIII and Fig. 9 summarise the intervention probe for AIPA (full). Removing the STI pathway disrupts the top-10 list far more than removing LTP (overlap with the original list 0.07–0.22 versus 0.62–0.72 on natural turns), and the per-turn driver label is *STI-driven* on 57–72% of natural turns and *Jointly-driven* on the rest; *LTP-driven* turns are below 1% and no turn is *Neither-driven*. On synthetic conflicts and overrides the STI dependence is strongest (overlap without STI 0.03–0.04, 81–84% STI-driven), and on synthetic *Uncertain* turns, where the intent cue has been erased, the label flips to *Jointly-driven* (81%), which is the behaviour one would want. Agreement between the driver label and the chosen action is 0.37 on natural turns and 0.43 on synthetic turns, lower than in the reduced-budget run because the model now fuses more often while the ranking is still intent-led. The diagnostic thus confirms what the ranking tables already suggest: the trained model leans heavily on the conversation and uses the profile

TABLE VI
RELATIONSHIP CLASSIFICATION, MEAN OVER SEEDS. W-F1 IS WEIGHTED F1; THE LAST FIVE COLUMNS ARE PER-CLASS F1 (–: CLASS ABSENT FROM THE SUBSET).

Model	Subset	Acc.	Macro-F1	W-F1	Compl.	Consist.	Confl.	Overr.	Uncert.
AIPA w/o relationship	natural	0.794	0.621	0.791	0.521	0.845	0.422	0.363	0.953
AIPA w/o relationship	synthetic	0.971	0.970	0.970	0.941	0.970	0.943	0.996	1.000
AIPA w/o counterfactual	natural	0.795	0.621	0.792	0.534	0.847	0.406	0.368	0.951
AIPA w/o counterfactual	synthetic	0.973	0.973	0.973	0.947	0.970	0.951	0.997	1.000
AIPA w/o clarification	natural	0.801	0.617	0.796	0.549	0.847	0.400	0.334	0.956
AIPA w/o clarification	synthetic	0.965	0.964	0.965	0.934	0.968	0.929	0.992	1.000
AIPA w/o persistence	natural	0.807	0.656	0.804	0.556	0.854	0.444	0.471	0.954
AIPA w/o persistence	synthetic	0.963	0.962	0.962	0.927	0.961	0.931	0.988	1.000
AIPA (rule policy)	natural	0.802	0.636	0.799	0.543	0.852	0.427	0.406	0.953
AIPA (rule policy)	synthetic	0.967	0.966	0.967	0.936	0.968	0.930	0.996	1.000
AIPA (full)	natural	0.800	0.627	0.797	0.553	0.848	0.412	0.369	0.954
AIPA (full)	synthetic	0.963	0.962	0.962	0.927	0.961	0.931	0.988	1.000

TABLE VII
ARBITRATION AND CLARIFICATION BEHAVIOUR, MEAN OVER SEEDS. CLARIFICATION PRECISION IS UNDEFINED WHEN NO QUESTION IS ASKED.

Model	Subset	Arb. acc.	Confl. res.	Overr. succ.	Clar. rate	Clar. prec.	Unnec. clar.	Wrong overr.
AIPA w/o relationship	natural	0.894	0.418	0.200	0.196	0.989	0.002	0.017
AIPA w/o relationship	synthetic	0.980	0.958	0.205	0.226	1.000	0.000	0.000
AIPA w/o counterfactual	natural	0.892	0.414	0.218	0.199	0.979	0.004	0.014
AIPA w/o counterfactual	synthetic	0.984	0.963	0.209	0.226	1.000	0.000	0.000
AIPA w/o clarification	natural	0.709	0.351	0.306	0.000	–	0.000	0.010
AIPA w/o clarification	synthetic	0.750	0.937	0.203	0.000	–	0.000	0.000
AIPA w/o persistence	natural	0.905	0.406	0.196	0.196	0.991	0.002	0.011
AIPA w/o persistence	synthetic	0.974	0.938	0.208	0.226	1.000	0.000	0.000
AIPA (rule policy)	natural	0.811	0.286	0.393	0.323	0.642	0.116	0.009
AIPA (rule policy)	synthetic	0.952	0.900	0.214	0.257	0.880	0.031	0.000
AIPA (full)	natural	0.902	0.353	0.263	0.196	0.991	0.002	0.010
AIPA (full)	synthetic	0.974	0.938	0.208	0.226	1.000	0.000	0.000

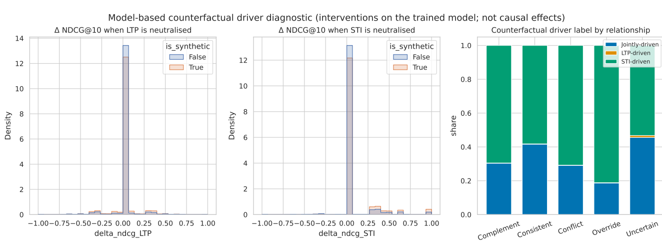

Fig. 9. Driver-label distribution and NDCG changes under signal removal.

as a secondary signal.

E. Persistent shifts

With $k = 2$ the tracker flagged 32 persistent shifts per seed for AIPA (full) across about 20 seekers (e.g. seeker 1087 repeatedly steering toward war films and horror, seekers 1034 and 1035 toward horror and romance). Because a shift is only applied to instances that come *after* the second qualifying session, it touched 2,321 test instances from 30 seekers, on which Hit@10 moved from 0.1166 without the tracker to 0.1172 with it (+0.0006, permutation $p = 1$); over the whole natural set the difference is +0.0003, and restricting to the 51 seekers with at least three sessions does not change the picture. A sweep over $k \in \{1, 2, 3\}$ on the validation split gives 168, 7 and 0.2 shifts per seed and no change in validation Hit@10 at any k; on the test split $k = 1$ fires 183 times and is, if anything, marginally worse. The mechanism works as designed and is now exercised often enough to be evaluated, and the evaluation says it does not move the ranking; the criterion that the tracker improves the instances it affects is not met.

F. Sensitivity and fusion weight

Figure 10 plots Hit@10 against history length, number of seeker turns and injected conflict intensity. Cold seekers are the easiest bucket (0.133), short histories the hardest (0.089 for 3–9 items), and the mid and long buckets sit in between (0.124 and 0.116); the same shape holds for every model, so it reflects the data rather than the arbitration. On synthetic conflicts, AIPA (full) reaches 0.23 at the mildest intensity (STI-only 0.24), 0.31 at the middle one (0.30) and 0.20 at the strongest (0.17); with 68–85 instances per cell none of these gaps is meaningful. Table IX shows the fixed-weight sweep for naive fusion: anything between $\alpha_{\mathrm{LTP}} = 0$ and 0.5 is close to the best (0.119–0.121), 0.75 already costs two points, and pure LTP weighting collapses to 0.031, which sets the scale for how much the profile alone can achieve here.

TABLE VIII
COUNTERFACTUAL PROBE OF AIPA (FULL) BY RELATIONSHIP CLASS: MEAN ABSOLUTE CHANGE IN NDCG@10 AND TOP-10 OVERLAP WHEN ONE PATHWAY IS REMOVED, AND THE SHARE OF TURNS ASSIGNED TO EACH DRIVER LABEL. THESE DESCRIBE THE MODEL, NOT THE USERS.

			|ΔNDCG@10|		Top-10 overlap		Driver (%)			
Source	Relationship	n	no LTP	no STI	no LTP	no STI	STI	LTP	Joint	Neither
natural	Complement	744	0.047	0.074	0.635	0.102	67	0	32	0
natural	Conflict	292	0.045	0.072	0.649	0.072	67	0	33	0
natural	Consistent	2070	0.035	0.055	0.619	0.176	58	0	41	0
natural	Override	39	0.048	0.087	0.692	0.092	72	1	28	0
natural	Uncertain	1327	0.018	0.049	0.715	0.216	57	1	42	0
synthetic	Complement	134	0.055	0.140	0.685	0.084	81	0	19	0
synthetic	Conflict	112	0.071	0.145	0.727	0.030	81	0	19	0
synthetic	Consistent	123	0.036	0.070	0.598	0.211	53	0	47	0
synthetic	Override	114	0.048	0.099	0.741	0.042	84	0	16	0
synthetic	Uncertain	141	0.028	0.029	0.520	0.380	15	4	81	0

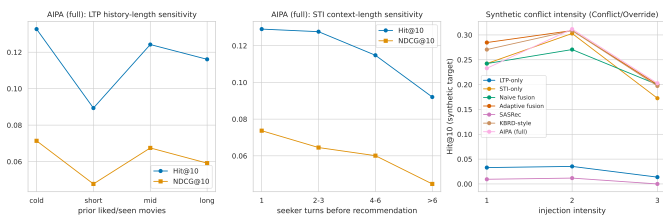

Fig. 10. Sensitivity of Hit@10 to history length, STI length and synthetic conflict intensity.

TABLE IX
FIXED FUSION WEIGHT SWEEP (NAIVE FUSION, NATURAL TEST SET UNLESS STATED).

α_{LTP}	Hit@10	NDCG@10	Hit@20	Hit@10 (synthetic)
0.00	0.119	0.062	0.186	0.172
0.25	0.121	0.064	0.191	0.177
0.50	0.119	0.063	0.191	0.176
0.75	0.100	0.052	0.158	0.152
1.00	0.031	0.016	0.049	0.024

G. Ablations and efficiency

Reading the AIPA rows of Tables III and IV as an ablation study, none of the removals changes natural Hit@10 by more than 0.003 and none is significant ($p_{\mathrm{Holm}} \geq 0.77$). Removing the relationship head, the counterfactual features or the persistence tracker leaves the score unchanged to three decimals; removing clarification is numerically the best variant (0.118) and the rule policy the best of all (0.120), neither significantly. On the strict conflict subset the signs are the ones one would hope for (removing persistence or the rule policy costs 0.005–0.006) but the effects are far inside the noise. All variants have 0.51–0.63 M parameters, train in under a minute on CPU (SASRec excepted) and score a turn in under 0.1 ms (Table X); the arbitration machinery costs roughly 5% extra parameters over naive fusion and about twice its training time, both negligible in absolute terms.

H. Qualitative cases and error analysis

Table XI shows representative test turns. The pattern is consistent with the aggregate numbers: when a seeker states a genre plainly ("inspirational war movies like PT 109"), the model reads the departure from a comedy-heavy profile as a conflict, prioritises STI and ranks the target seventh; when the request carries no genre cue ("anything recent that you have seen?"), the model asks, and the target sits far down the list. The most instructive failure is negation: "I like pretty much anything but horror!" yields a horror-dominated STI vector, the weak rule calls it an override, and the model follows the (wrong) intent to rank 144. The most common error type overall is a weak-rule *Complement* or *Conflict* turn predicted as *Consistent* and fused; in the natural set 44% of *Complement* and 64% of *Conflict* references are mispredicted (down from 75% and 81% in the reduced-budget run), and their targets have median ranks of 165 and 144 against 186 for *Consistent* turns.

TABLE X
MODEL SIZE AND COST (CPU ONLY).

Model	Parameters	Size (MB)	Train (s)	CPU inf. (ms/sample)
LTP-only	597,525	2.39	19.0	0.032
STI-only	597,525	2.39	24.1	0.045
Naive fusion	597,525	2.39	21.0	0.093
Adaptive fusion	608,790	2.44	24.7	0.056
Sequential (GRU)	513,357	2.05	49.6	0.052
Conversation-aware	537,613	2.15	12.1	0.010
SASRec	546,893	2.19	355.8	0.067
KBRD-style	539,254	2.16	14.1	0.013
AIPA w/o relationship	629,150	2.52	37.3	0.048
AIPA w/o counterfactual	629,150	2.52	35.3	0.041
AIPA w/o clarification	629,470	2.52	38.5	0.043
AIPA w/o persistence	629,470	2.52	42.6	0.054
AIPA (rule policy)	617,306	2.47	42.0	0.062
AIPA (full)	629,470	2.52	53.1	0.071

VIII. DISCUSSION

A. What the evidence supports

Four things are supported by this run. (i) Any model that reads the conversation beats a profile-only model on ReDial by a wide, significant margin; the profile alone, whether pooled (LTP-only) or modelled sequentially (GRU, SASRec), reaches 5–8% Hit@10 against 12–13% for the intent-reading models. (ii) The learned arbitration policy is well-behaved: it asks when it should, almost never asks when it should not, and almost never overrides in the wrong direction; and with the full corpus the relationship head has become a usable classifier (macro-F1

TABLE XI
SELECTED TEST TURNS (N: NATURAL, S: SYNTHETIC). REFERENCE LABELS ARE WEAK-RULE LABELS; "RANK" IS THE RANK OF THE TRUE TARGET UNDER AIPA (FULL).

	Dialogue excerpt (seeker turns before the target)	LTP profile	STI signal	Ref./pred. relationship	Action	$w_{\mathrm{LTP}}/w_{\mathrm{STI}}$	Driver	Rank
N	Recommender: Hello ∣ Recommender: What kind of movies are you into ?! ∣ Seeker: Hi! How about inspirational war movies like PT 109 (1963) or USS Indianapolis: Men of...	Horror 0.35; Drama 0.23; Thriller 0.15	War 0.75; Action 0.12; Drama 0.12	Complement / Conflict	Prioritize_STI	0.43/0.57	STI	7
N	Recommender: Good morning. Have any plans for the weekend? ∣ Seeker: Good morning. ∣ Seeker: Yes, to watch a bunch of animated movies! Like The Incredibles (2004) to g...	Comedy 0.24; Animation 0.22; Children 0.17	Animation 0.61; Action 0.11; Adventure 0.11	Consistent / Consistent	Fuse	0.49/0.51	Jointly	3
N	Recommender: One of my favorites is The Jerk (1979). Have you seen it? ∣ Seeker: I've never seen it ∣ Recommender: It's a bit older. Tropic Thunder (2008) is more rece...	Drama 0.28; Action 0.19; Adventure 0.10	Comedy 1.00	Conflict / Consistent	Fuse	0.48/0.52	STI	6
N	Seeker: Hows it going? ∣ Recommender: Good. ∣ Seeker: I like pretty much anything but horror! ∣ Seeker: Anything recent that you have seen?	Comedy 0.18; Mystery 0.14; Crime 0.11	Horror 1.00	Override / Consistent	Prioritize_STI	0.43/0.57	STI	144
N	Seeker: i am in the mood to watch movies ∣ Recommender: Hi What kind of movies are you into ∣ Seeker: anything you would like to recommend? ∣ Seeker: I am open to any ...	Comedy 0.65; Romance 0.14; Crime 0.06	(no genre cue)	Uncertain / Uncertain	Ask_Clarification	0.45/0.55	Jointly	639
N	Recommender: Of course I have to mention The Bodyguard (1992) ∣ Recommender: Awww. I thought that one was great. ∣ Seeker: Oh, yes! With Whitney Houston. That's one ...	Comedy 0.44; Horror 0.17; Drama 0.12	Action 0.17; Adventure 0.17; Children 0.17	Complement / Complement	Fuse	0.48/0.52	Jointly	6167
N	Recommender: Hello. How are you? ∣ Seeker: Hey there, I'm looking for a good thriller movie. Do you know any good ones?	(none: cold seeker)	Thriller 1.00	Uncertain / Uncertain	Prioritize_STI	0.41/0.59	STI	275
N	Recommender: What kind of movies do you like? ∣ Seeker: Hello ∣ Seeker: i open to any movie	Action 0.28; Sci-Fi 0.28; Adventure 0.21	(no genre cue)	Uncertain / Uncertain	Ask_Clarification	0.45/0.55	Jointly	44

0.63, Conflict-F1 0.41) rather than a majority-class predictor. (iii) On plainly stated departures from habit (the synthetic set), AIPA follows the user, and the counterfactual probe shows that the ranking really is driven by the stated intent in those cases. (iv) The whole apparatus is cheap: the arbitration components add about 5% parameters and no measurable inference cost.

B. What it does not support

The central claim, that explicit arbitration improves recommendations over simpler fusion *specifically on conflict turns*, is not supported, and with the full corpus this is no longer a matter of statistical power. On the strict natural conflict subset AIPA is two points behind the intent-only and KBRD-style models with overlapping intervals; on the broad subset the gap is one and a half points and significant; on synthetic conflicts AIPA ties adaptive fusion and the trivial intent-following policy. On the full natural set the intent-only and KBRD-style models are about one point better, significantly. The ablations are flat: removing any single arbitration component changes nothing, which means the components are not what determines the ranking. We see three reasons, in rough order of importance.

First, the labels. Weak-rule relationship labels are the reference for the relationship head, the arbitration metrics and the conflict subset itself. If the rule mislabels many turns, we are training toward noise and evaluating on a subset that is not what we think it is. The negation example in Table XI is one concrete instance; the near-perfect synthetic results versus the moderate natural results point the same way.

Second, the signal. The counterfactual probe says the trained model is intent-driven on 60–70% of natural turns and profile-driven on fewer than 1%. When the ranking is already dominated by one pathway, an arbitration layer that decides how much to trust the other pathway has little to arbitrate, and the flat ablations follow. ReDial seekers are crowd workers who discuss a handful of films per session; their cross-session "profile" is short, noisy, and only weakly predictive (5% Hit@10 alone). A corpus with genuine long-term histories is needed before arbitration can pay off.

Third, the representation. MiniLM sentence embeddings improved every model over the earlier TF–IDF run (STI-only $0.110 \to 0.125$, AIPA $0.095 \to 0.117$) but did not change the ordering. The STI encoder still reduces an utterance to genre proportions plus a pooled sentence vector, and cues such as "for my brother" or "anything but horror" are not recovered; the *Uncertain* class is 30% of natural turns and is where the strongest baselines gain their edge.

C. What a conclusive test needs

1) A corpus with genuine long-term histories. ReDial's cross-session profiles come from crowd workers who each completed a few tasks; a profile that alone reaches 5% Hit@10 gives an arbitration layer nothing to weigh against the intent. Amazon-review-grounded dialogue corpora or a platform log with real return visits would be the natural next testbed.
2) Human relationship annotations for a few hundred test turns. The pipeline already accepts such a file and reports all metrics against it separately; this is the single most valuable addition, and it would settle whether the weak rule is the reason *Conflict*-F1 stays at 0.41.

3) An STI encoder that handles negation and pragmatics. "Anything but horror" must not become a horror intent; a small fine-tuned classifier over the sentence embedding, or an instruction-tuned language model prompted for genre preferences with polarity, would address the most visible failure mode.
4) A faithful knowledge-grounded baseline (KBRD/KGSF with their knowledge graph), since the KBRD-style re-implementation here already beats AIPA and a positive result would have to hold against the original models.
5) Evaluation of clarification with users, or at least with simulated answers; offline precision says nothing about whether a question was welcome, and a question that is answered is the one place where arbitration can still add information that no single-shot baseline has.

D. Threats to validity

Conversation identifiers are assumed chronological; ReDial seekers are crowd workers, not organic users, so cross-session "preferences" may partly reflect task instructions; the MovieLens join misses a fraction of titles, leaving those items without genres; and the sequential, SASRec, conversation-aware and KBRD-style baselines are our own small re-implementations. All of these are documented in the released code and reported in the generated experiment report.

IX. Conclusion

We set out to make the reconciliation of long-term preference and short-term intent explicit in a conversational recommender and to test whether doing so helps where it should, on turns where the two disagree. We built a complete, reproducible pipeline on an English corpus, with leak-free cross-session histories, two clearly separated label sources covering all five relationship classes, a relationship classifier trained with conflict-weighted and self-trained weak labels, a learned arbitration policy with learned fusion weights and clarification, a persistence tracker and a counterfactual probe, and we evaluated it on the full corpus with five seeds against eight baselines with intervals, corrected paired tests and pre-stated success criteria. The result is clear and negative on the ranking question: the arbitration components behave sensibly and make the model's decisions legible, but on ReDial they do not improve recommendations over an intent-only model or a KBRD-style baseline, on conflict turns or elsewhere, and removing them changes nothing. We have not hidden that. What the study does deliver is a tested framework, an honest measurement protocol for the conflict case, and a diagnosis: the profile signal in this corpus is too weak for arbitration to matter. The next step is not another model component but a corpus with real long-term histories and human relationship labels; the code is ready for both.

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